

MD. RAKIB MAHMUD HRIDOY

Android Engineer · Mobile Application Developer

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PROFESSIONAL SUMMARY

Mobile engineer with 5+ years building and shipping production applications in Kotlin, Jetpack Compose, and Flutter. Experienced across real-time communication systems — SIP/VoIP telephony over Asterisk and self-hosted video conferencing infrastructure — and in delivering feature modules within Clean Architecture and BLoC codebases. Comfortable owning features end to end, from API integration through Play Store release.

TECHNICAL SKILLS

Native Android: Kotlin, Java, Jetpack Compose, Coroutines & Flow, Hilt, Room, Retrofit, Material Design, Google Play Billing

Architecture: MVVM, MVI, Clean Architecture, multi-module design, dependency injection, offline-first patterns

Flutter: Dart, BLoC / flutter_bloc, GetIt, GoRouter, Drift, Dio, Firebase Crashlytics & Remote Config

Other Cross-Platform: React Native, JavaScript, TypeScript

Real-Time & Telephony: SIP, VoIP, Asterisk AMI (asterisk-java), Jitsi stack (Prosody, Jicofo, JVB, Coturn), WebRTC

Backend & Data: RESTful API design, MySQL, SQLite, ORMLite + SQLCipher, Firebase services, JWT authentication

Tooling & Practice: Git, Docker, Azure, Nginx, Jira, Agile/Scrum, Play Store release management, AI-assisted development (Claude Code, Antigravity, Gemini)

PROFESSIONAL EXPERIENCE

Software Engineer II | Brotecs Technologies Ltd, Dhaka, Bangladesh

December 2020 – July 2026

- Architected and migrated the flagship Android codebase to a multi-module structure, cutting incremental build times and allowing the team to work in parallel without merge conflicts.
- Shipped and maintained 5 production Android applications through full Play Store release cycles, including Google Play Billing integration, staged rollouts, and post-release crash triage.
- Led adoption of Jetpack Compose, migrating screens from legacy XML views and standardising a shared design-system component library used by the whole mobile team.
- Owned client-facing requirements gathering for enterprise telephony clients, translating operational needs into technical specifications and delivery timelines.
- Collaborated with design, QA, product, and offshore engineering teams in an Agile/Scrum process, contributing to code review and release planning across time zones.
- Introduced agentic AI tooling (Claude Code, Antigravity) into the team's development workflow for large-scale refactoring and debugging.

SELECTED PROJECTS

Employee HR & Attendance Platform — *Flutter, Dart, BLoC, Drift, REST*

- Built the attendance and time-tracking module end to end — clock in/out, live punch status, and month-by-month attendance history — for an HR platform.
- Implemented manager approval workflows including bulk approve/reject, rejection-reason capture, and status filtering for early-leave and extra-time requests.
- Delivered all features within an existing Clean Architecture and BLoC codebase, following per-feature data/domain/presentation layering with GetIt dependency injection and GoRouter navigation.

Cabin Crew Calling Console — *Android, Java, Asterisk AMI, SIP, Ormlite + SQLCipher*

- Replaced an aircraft's built-in hardware interphone console with an Android application, giving cabin crew the same call answering, forwarding, and cabin-extension routing on standard devices.
- Integrated asterisk-java to subscribe to the Asterisk AMI event stream and reconstruct live state for every extension and trunk — in-call, multi-call, and trunk usage — keeping the model in sync as events arrived so operators saw current status rather than stale data.
- Built the entire application interface, with call records and configuration stored locally in Ormlite behind SQLCipher encryption.
- Used Firebase Remote Config for configuration changes and forced updates, removing manual redeployment to each installation.

VoIP Calling Applications — *Android, Java, SIP, in-house SDK, OrmLite + SQLCipher, Firebase*

- Built softphone applications for organisations running their own telephony servers — internal extension-to-extension calling with outside calls routed through a configured trunk — deployable against any SIP server.
- Owned integration of the company's in-house communication SDK into the call-handling flow, ensuring placing, receiving, and routing behaved consistently in every direction.
- Deployed in aviation, carrying aircraft-to-ground, ground-to-aircraft, and aircraft-to-aircraft calling alongside internal cabin extensions.

Virtual Second-Number Application — *Android, Java, Kotlin, SIP, REST, Google Play Billing*

- Shipped a Google Play consumer application selling SIP-based virtual phone numbers, letting users hold an additional international number alongside their existing SIM and call internationally from it.
- Integrated external provider APIs through the cloud backend to source and manage number inventory, making numbers browsable and purchasable in-app.
- Implemented payments and subscriptions with Google Play Billing, handling failed payments and subscription state changes rather than assuming successful purchase.
- Built the registration and verification flow for jurisdictions requiring identifying information before a number can be issued, with explicit guidance on missing requirements.

Enterprise Video Conferencing Platform — *Jitsi, Prosody, Jicofo, JVB, Coturn, Debian, Nginx*

- Delivered a licensed, organisation-based conferencing platform with a CMS for administrators to manage members, invite external participants, and schedule sessions.
- Took over the server deployment after an initial Ubuntu-based attempt stalled, provisioning and configuring the full Jitsi stack on Debian behind an Nginx reverse proxy until the system was operational.
- Hardened the deployment for production: JWT authentication, Prosody domain matching, Jicofo trusted-domain configuration, secure WebSocket upgrade, and firewall rules.
- Worked with colleagues on the web and mobile clients, which carry session recording and an in-meeting whiteboard for drawing, PDF presentation, and slide sharing.
- Built a plugin embedding the platform inside Moodle LMS, letting course users create and join scheduled sessions without leaving Moodle.

Telemedicine Consultation Application — *React Native, REST*

- Built the interface and API layer for a clinical application working with Mintti Vision, Heartbook ECG, and digital stethoscope devices, wiring the UI to backend services so completed measurements travel from device to server and can be retrieved elsewhere.
- Implemented consultation booking, allowing patients to schedule doctor sessions against their recorded measurements. Internal SaaS product; scope was UI and API integration.

EDUCATION

B.Sc. in Computer Science and Engineering

Bangladesh University of Business and Technology (BUBT), Dhaka · February 2020 · CGPA 3.82 / 4.00

CERTIFICATIONS & TRAINING

- **Flutter Development Bootcamp with Dart** — Udemy, 2026
- **Mobile Application Development (Android)** — BITM, BASIS Institute of Technology and Management, 3 months, 2019
- **UI/UX Design** — ICT Ministry of Bangladesh, 3 months, 2018